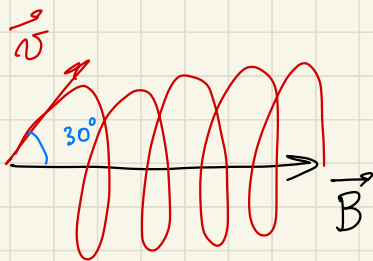


(19)



Alternative: $v_{\perp} = v \sin 30^{\circ}$ (+1)

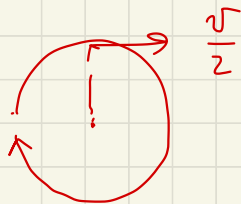
$$\frac{mv_{\perp}^2}{R} = qv_{\perp}B \Rightarrow$$

$$R = \frac{mv_{\perp}^2}{qv_{\perp}B} = \frac{mv_{\perp}}{qB} \quad (+1)$$

(18) Circumference = $2\pi R$ $\Rightarrow$ (+1)

(a) for the transverse view

$$v_{\perp} = v \sin 30^{\circ} = \frac{v}{2} \quad (+1)$$



$$\frac{m \left(\frac{v}{2}\right)^2}{R} = q \left(\frac{v}{2}\right) B \Rightarrow$$

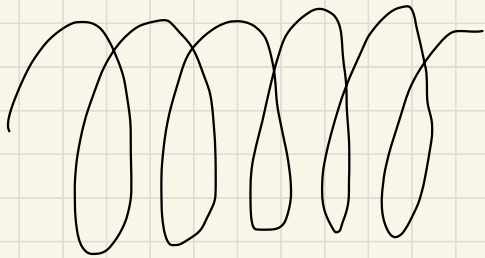
$$\frac{m \frac{v}{2}}{R} = qB \Rightarrow$$

$$R = \frac{mv}{2qB} = \frac{1.7 \cdot 10^{-27} \text{ kg} \cdot 3 \cdot 10^6 \text{ m/s}}{2 \cdot 1.6 \cdot 10^{-19} \text{ C} \cdot 0.1 \text{ T}} =$$

$$= 0.159 \text{ m}$$

$$T = \frac{2\pi R}{v_{\perp}} = \frac{4\pi R}{v} = 6.7 \times 10^{-7} \text{ s} \quad (+1)$$

(c)



Any effort on sketching
"believable" helix warrants (+3)

(d) $v_{||}$ does not change as the force projection onto $\vec{B}$ is zero.
 $\vec{F} = q\vec{v} \times \vec{B}$, so $\vec{F} \perp \vec{B} \Rightarrow$ no acceleration along $\vec{B}$. (+1)

$$v_{||} = v \cos 30^\circ = v \frac{\sqrt{3}}{2} \quad (+1)$$

$$\begin{aligned} \text{distance} &= v_{||} \cdot T = v \frac{\sqrt{3}}{2} \cdot 6.7 \cdot 10^{-7} \text{ s} = 3 \cdot 10^6 \text{ m/s} \cdot \frac{\sqrt{3}}{2} \cdot 6.7 \cdot 10^{-7} \text{ s} = \\ &= 1.74 \text{ m} \quad (+1) \end{aligned}$$

(20) (a) Magnetic field has to be perpendicular to the plane of moving current:

$$d\vec{B} = \frac{\mu_0 i}{4\pi} \frac{d\vec{\ell} \times (\vec{r} - \vec{s})}{|\vec{r} - \vec{s}|^3}$$

because of $d\vec{B} \sim d\vec{\ell} \times \dots$, and current flows along $d\vec{\ell}$.

But the cable is rotationally symmetric, so $\vec{B} = 0$ at $r=0$.

To graders $\vec{B} = 0$ must be justified. Acceptable solution is symmetry (explained, not just because of symmetry). If they use $\oint \vec{B} d\vec{s} = \mu_0 i_{enc} \rightarrow$ a limit must be taken, as loop does not exist for $r=0$. If they say $i_{enc} = 0$ for $r=0$ give at most 1 point.

(b) For (b), (c), (d) use Ampere's law: $\oint \vec{B} d\vec{\ell} = \mu_0 i_{enc} \Rightarrow$

(b)

$$\begin{aligned} (+1) \quad 2\pi r B &= \mu_0 i_{enc} \Rightarrow \\ B &= \frac{\mu_0 i}{2\pi r} = \frac{2 \cdot 10^{-7} \frac{T \cdot m}{A} \cdot 2A}{2 \cdot 0.5m} = 8 \cdot 10^{-7} T \quad \text{CCW} \end{aligned}$$

(c)

Out of page current of 2A in the core, in the page has to be calculated. The current density in the shield is

$$j = \frac{i}{\text{Surface area}} = \frac{2A}{\pi(1.2m)^2 - \pi(1m)^2} \quad (+1)$$

$$\text{So in the page current} = j \cdot (\pi(1.1m)^2 - \pi(1m)^2) \Rightarrow$$

$$i_{in} = \frac{2A \cdot (\pi(1.1m)^2 - \pi(1m)^2)}{\pi(1.2m)^2 - \pi(1m)^2} = 2A \cdot 0.477 \approx 0.95A \quad (+1)$$

$$B = \frac{\mu_0}{2\pi r} \cdot (2A - 0.95A) = 1.91 \cdot 10^{-7} T \quad (+1)$$

(1)

$$B = \frac{\mu_0 i}{2\pi r} = 0 \quad \text{as} \quad i_{\text{enc}} = 0 \text{ A at } r > 1.2 \text{ m.}$$

(21) (a) Use RHR to determine $\vec{B}$ from the inner solenoid $\rightarrow$ out of the page. From the outer solenoid $\rightarrow$ into. (+1)

$$B = \mu_0 n i \quad (+1)$$

$$\text{So } B_{\text{center}} = \mu_0 n_S i_S - \mu_0 n_L i_L = \mu_0 \left(2000 \frac{\text{turns}}{\text{m}} \cdot 1 \text{ A} - 4000 \frac{\text{turns}}{\text{m}} \cdot 1 \text{ A} \right) \\ = -2.5 \text{ mT} \quad (\text{abs value for magnitude, } 2.5 \cdot 10^{-3} \text{ T}). \quad (+1)$$

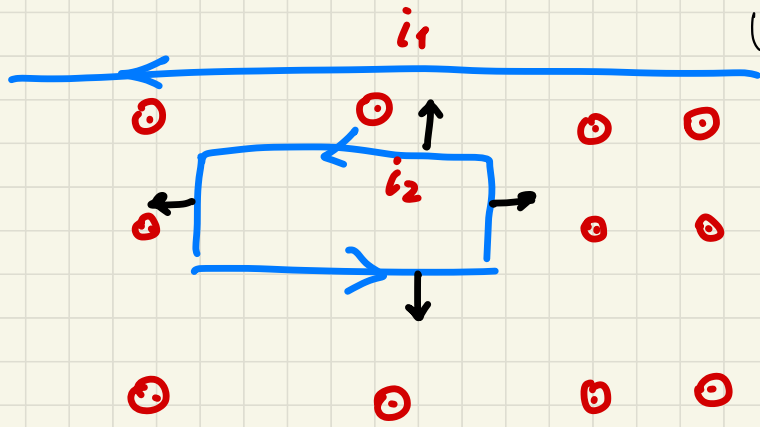
Direction is **into** the page

(b) No B from inner solenoid, so $B = \mu_0 n_L i_L = 5 \cdot 10^{-3} \text{ T}$ (+1)

(c) $\mu = 100$ indicates that the answer in (a) needs to be multiplied by 100, re.: 0.25 T

(22) (a) $\vec{F} = i \vec{L} \times \vec{B}$

Use RHR $\rightarrow \vec{B}$ field is out of the page in the area of the loop:



Vertical Sides will cancel out, but $\vec{B}$ at top side is stronger than at the bottom, so there will be an overall attractive force:

$$F = i_2 B_{\text{top}} W - i_2 B_{\text{bottom}} W$$

$$F = i_2 W (B_{\text{top}} - B_{\text{bottom}}) \quad (+2)$$

$$B_{\text{top}} = \frac{\mu_0 i}{2\pi D} \quad \text{from Ampere's law} \quad (+1)$$

$$F = \frac{i_2 W \mu_0 i_1}{2\pi} \left(\frac{1}{D} - \frac{1}{D+H} \right) = \frac{\mu_0 i_1 i_2 W H}{2\pi (D+H) D} \quad (+1)$$

(b)

$$\underbrace{\mu_D}_{(+1)} = iS = i_2 \underbrace{WH}_{(+1)}$$

(c)

$$\vec{\tau} = \vec{\mu} \times \vec{B} \quad (+1) \quad \text{The direction of } \vec{\mu} \text{ is out of page (RHR)} \quad (+1)$$

So is $\vec{B} \Rightarrow \vec{\mu}$ and $\vec{B}$ are collinear, so $\vec{\tau} = 0 \quad (+1)$